

Formal dependency draft for the vanishing critical 2-variation prescribed-angle rectangle claim

Issue #134; companion specification for PR #122

Draft of 30 August 2026

Abstract

This document proves the prescribed-angle rectangle theorem for Jordan parametrizations of vanishing critical 2-variation. It gives the finite-partition, regular-smoothing, excursion, winding, and Liouville-primitive arguments and pins every external theorem to the exact version and location independently checked. The Asano–Ike definition and off-diagonal conclusion supply four distinct vertices, so the $\theta = \pi/2$ case is a positive-size square.

1 Status vocabulary and theorem interface

Every lemma below carries one of the following statuses.

- **PROVED:** proved in this document from earlier displayed definitions and stated standard facts.
- **CITED:** imported from a named external source; citation fidelity remains part of the verification obligation.
- **STANDARD-CITED:** an exact, independently checked classical theorem is used but not reproved foundationally here.

The intended dependency chain is

$V_0 \longrightarrow$ polygonal approximation $\longrightarrow$ regular smoothing
 $\longrightarrow$ uniform local 2-variation $\longrightarrow$ primitive compactness $\longrightarrow$ Asano–Ike conclusion.

The internal arrows are proved below; external inputs are exact cited theorems whose hypotheses are explicitly discharged.

2 Vanishing 2-variation and polygonal error

Let $c: \mathbb{R} \rightarrow \mathbb{R}^2$ be continuous, 1-periodic, and injective on $[0, 1)$. For an interval $I = [a, b]$, define

$$\|x\|_{2\text{-var}; I}^2 := \sup_{a=t_0 < \dots < t_m = b} \sum_{j=0}^{m-1} |x(t_{j+1}) - x(t_j)|^2.$$

For a partition $D = \{0 = t_0 < \dots < t_m = 1\}$ put

$$\text{mesh}(D) = \max_j (t_{j+1} - t_j), \quad w_2(c, \delta)^2 := \sup_{\text{mesh}(D) \leq \delta} \sum_j |c(t_{j+1}) - c(t_j)|^2.$$

Both suprema initially take values in $[0, \infty]$.

Definition 1 (Critical vanishing-variation hypothesis). Write $V_0(c)$ for

$$\exists \delta_0 > 0 : w_2(c, \delta_0) < \infty \quad \text{and} \quad w_2(c, \delta) \longrightarrow 0 \quad (\delta \downarrow 0).$$

Thus every real-valued variation calculation below occurs in the finite regime $0 < \delta \leq \delta_0$.

Lemma 2 (Lifted short-interval modulus). *If $V_0(c)$ holds, $0 < \delta \leq \frac{1}{2}$, and $I \subset \mathbb{R}$ has $|I| \leq \delta$, then*

$$\|c\|_{2\text{-var}; I} \leq \sqrt{2} w_2(c, \delta).$$

Status: PROVED.

Proof. If I does not cross an integer, extend any partition of I to a full-period partition of mesh at most δ . If it crosses one integer, insert the seam. The insertion replaces at most one increment $u + v$ by u, v , and $|u + v|^2 \leq 2(|u|^2 + |v|^2)$. After reducing the two pieces modulo one, fill every complementary gap by subintervals of length at most δ and combine the nodes into a full-period partition. Since $|I| \leq \delta \leq 1/2$, at most one integer seam occurs. Take suprema. $\square$

For a partition D , let c^D be the affine interpolant of c on every interval of D .

Lemma 3 (Every-partition interpolation estimate). *For every partition D of $[0, 1]$,*

$$\|c - c^D\|_{2\text{-var}; [0,1]} \leq \sqrt{8} w_2(c, \text{mesh}(D)). \quad (2.1)$$

Consequently $V_0(c)$ implies $c^D \rightarrow c$ in 2-variation along every sequence with $\text{mesh}(D) \rightarrow 0$.

Status: PROVED; replaces the load-bearing Yang quantifier.

Proof. Write $I_j = [t_j, t_{j+1}]$ and $y = c - c^D$. Since y vanishes at every t_j , inserting all partition nodes into an arbitrary variation partition costs at most a factor 2 in the squared sum. Indeed, although one crossing increment may meet many nodes of D , $y = 0$ at every inserted node, so only its two endpoint pieces can be nonzero. Hence

$$\|y\|_{2\text{-var}}^2 \leq 2 \sum_j \|y\|_{2\text{-var}; I_j}^2.$$

On I_j , Minkowski's inequality and the fact that the affine chord has 2-variation equal to $|c(t_{j+1}) - c(t_j)|$ give

$$\|y\|_{2\text{-var}; I_j} \leq 2 \|c\|_{2\text{-var}; I_j}.$$

Near-optimal local partitions concatenate into one partition of mesh at most $\text{mesh}(D)$, so

$$\sum_j \|c\|_{2\text{-var}; I_j}^2 \leq w_2(c, \text{mesh}(D))^2.$$

Combining the three inequalities proves (2.1). $\square$

External interface 4 (Simple interpolation, BG). For every $\varepsilon > 0$ and every Jordan parametrization c , there is a partition D with $\text{mesh}(D) < \varepsilon$ such that the parametrized affine interpolant c^D is a simple polygonal Jordan parametrization.

Status: CITED: Boedihardjo–Geng, arXiv:1309.1576v2, Theorem 2.2, PDF pp. 6–7; the fixed-parameter, parameter-mesh, unique-minimizing-geodesic, and Jordan-interpolant clauses were independently checked. In $\mathbb{R}^2$ the geodesics are affine chords.

Under Interface 4, choose D_n with mesh tending to zero and put $P_n = c^{D_n}$. The preceding proved lemma gives

$$\|P_n - c\|_{2\text{-var}} \rightarrow 0, \quad P_n \rightarrow c \text{ uniformly}, \quad (2.2)$$

the uniform statement following because $P_n(0) = c(0)$.

3 Regular parametrized smoothing

Lemma 5 (Regular parametrized corner smoothing). *For every parametrized simple polygon $P: \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}^2$ and every $\varepsilon > 0$, there is a C^∞ regular Jordan embedding $Q: \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}^2$ such that*

$$\|Q - P\|_{1\text{-var}} < \varepsilon, \quad (3.1)$$

Here regular means $Q'(t) \neq 0$ at every parameter, including the periodic seam.

Status: PROVED.

Proof. Let $0 = t_0 < \dots < t_m = 1$ be the edge partition on a periodic lift, let v_i be the nonzero constant velocity on the i th edge, and put $M = \max_i |v_i|$. Choose an even, nonnegative, smooth periodic mollifier ρ_h , supported at circular distance less than h and having integral one, where $2h$ is smaller than every edge's parameter length. Set $Q_h = P * \rho_h$. Writing $Q_h(t) = \int_{\mathbb{R}/\mathbb{Z}} P(r) \rho_h(t - r) dr$, differentiation of the smooth kernel under the integral makes Q_h smooth. Periodic integration by parts, using absolute continuity of P , gives

$$Q'_h = P' * \rho_h. \quad (3.2)$$

Thus Q_h is periodic and smooth, including at $0 = 1$. Away from the h -neighborhoods of vertices, $Q'_h = P' = v_i$. Near the vertex between v_{i-1} and v_i , equation (3.2) is a convex combination $\alpha v_{i-1} + \beta v_i$, with $\alpha, \beta \geq 0$ and $\alpha + \beta = 1$.

The adjacent velocities cannot lie on opposite rays, since then the two incident polygonal edges would overlap near their common vertex. Hence a linear functional ℓ_i is strictly positive on both. It follows that $\ell_i(Q'_h) > 0$ throughout the vertex neighborhood. Therefore Q'_h never vanishes.

To keep the injectivity quantifiers uniform, first fix $h_0 < \frac{1}{4} \min_i (t_{i+1} - t_i)$ and a finite cover of $\mathbb{R}/\mathbb{Z}$ by vertex intervals of radius $2h_0$ and intervening edge-interior intervals. Shrink h_0 once, if necessary, so the positivity argument applies on each fixed vertex interval for every $0 < h < h_0$; on the remaining pieces Q'_h is an edge velocity. Hence Q_h is injective on every member of this one fixed cover for every $0 < h < h_0$.

Let $\eta > 0$ be a Lebesgue number for the fixed cover. On the compact set of parameter pairs at circular distance at least η , injectivity of P gives a positive minimum separation d_η . Moreover,

$$|Q_h(t) - P(t)| \leq \sup_{d_{\mathbb{R}/\mathbb{Z}}(s,0) < h} |P(t-s) - P(t)|,$$

which tends uniformly to zero. Choose h so that $\|Q_h - P\|_\infty < d_\eta/3$. Such pairs remain separated by Q_h ; pairs closer than η lie in a common cover member and are separated by local injectivity. Hence Q_h is a regular Jordan embedding.

Finally, $Q_h - P$ is absolutely continuous, so its 1-variation is the integral of $|Q'_h - P'|$. This integrand vanishes off the union U_h of the m vertex neighborhoods, while $|U_h| \leq 2mh$ and both derivatives have norm at most M there. Consequently

$$\|Q_h - P\|_{1\text{-var}} = \int_0^1 |Q'_h - P'| \leq 4mMh.$$

Taking also $h < \varepsilon/(4mM)$ proves (3.1). If orientation is part of the smooth-Jordan convention, it is preserved: choose a point in the bounded component of $\mathbb{R}^2 \setminus P(\mathbb{R}/\mathbb{Z})$ and then h so small that the straight homotopy from P to Q_h avoids it; winding-number homotopy invariance fixes the sign. $\square$

Using Lemma 5, choose c_n from P_n with errors $\varepsilon_n \downarrow 0$. Since $\|x\|_{2\text{-var}} \leq \|x\|_{1\text{-var}}$, (2.2) gives

$$\|c_n - c\|_{2\text{-var}} \rightarrow 0, \quad c_n \rightarrow c \text{ uniformly.} \quad (3.2)$$

4 Uniform local variation

Define, for $0 < \delta \leq \frac{1}{2}$,

$$\nu(\delta) = \sup_n \sup_{0 \leq t-s \leq \delta} \|c_n\|_{2\text{-var};[s,t]}, \quad (4.1)$$

where intervals are interpreted on the lifted periodic parameter.

Lemma 6 (Whole-family local modulus). *Under (3.2), $\nu(\delta) \rightarrow 0$ as $\delta \downarrow 0$.*

Status: *PROVED.*

Proof. Put $e_n = \|c_n - c\|_{2\text{-var};[0,1]}$. Restriction, the seam insertion argument, and Lemma 2 give

$$\|c_n\|_{2\text{-var};[s,t]} \leq \sqrt{2}(w_2(c, t-s) + e_n). \quad (4.2)$$

Given $\eta > 0$, first choose N so the right side is small for $n \geq N$, specifically $\sqrt{2}e_n < \eta/2$, then choose δ using $V_0(c)$ so $\sqrt{2}w_2(c, \delta) < \eta/2$. For the finite prefix $n < N$, each regular C^1 map is Lipschitz, say with constant L_n , and

$$\sum_i |\Delta c_n|^2 \leq (\max_i |\Delta c_n|) \sum_i |\Delta c_n| \leq L_n^2 \delta^2.$$

Taking the finite maximum proves the claim. $\square$

5 Winding control and primitive compactness

External interface 7 (Index-current identity). For every closed rectifiable planar curve L , the winding function is compactly supported and belongs to $L^1(\mathbb{R}^2)$, and the current with density $\text{Ind}(L, \cdot)$ has boundary equal to the integration current $[L]$.

Status: CITED: integrability from Carmona–Cufí, Theorem 2; boundary identity from the main theorem of Cufí–Verdera applied to smooth compactly supported test functions. Exact pins: J. J. Carmona–J. Cufí, J. Analyse Math. 120 (2013), 225–253, DOI 10.1007/s11854-013-0019-9, preprint Theorem 2, PDF p. 8; and arXiv:1306.6832v1, main theorem, PDF p. 1. Both were independently checked.

Compact support here is elementary: the trace of L lies in a ball, and winding number is zero on the unbounded exterior component. Carmona–Cufí gives L^2 ; finite measure of the supporting ball then gives L^1 .

Proposition 8 (Embedded-arc winding estimate). *Using the cited Interface 7, let $a: [u, v] \rightarrow \mathbb{R}^2$ be an injective rectifiable arc. Close it by the oriented chord from $a(v)$ to $a(u)$, obtaining L . Then*

$$\int_{\mathbb{R}^2} |\text{Ind}(L, z)| dz \leq \frac{\pi}{4} \|a\|_{2\text{-var};[u,v]}^2. \quad (5.1)$$

Status: *PROVED from the independently checked cited index-current, Jordan, and isodiametric theorems.*

Proof. Let H be its supporting line and S the closing chord. The relatively open set $\{t \in (u, v) : a(t) \notin S\}$ is a countable disjoint union of intervals (α_j, β_j) . Continuity puts both endpoint images on S . Injectivity makes them distinct, and the excursion interior misses S ; hence the excursion closed by its subchord is a Jordan loop E_j . Gilman–Kra–Rodríguez, Theorem 5.23, supplies its bounded complementary domain Ω_j and $\text{Ind}(E_j, \cdot) = \sigma_j \mathbf{1}_{\Omega_j}$ almost everywhere, with $\sigma_j \in \{\pm 1\}$.

Adding the subchord does not change the excursion diameter: if x is on the subchord and y is in the excursion closure, convexity of distance along the chord gives $|x - y| \leq \max(|a(\alpha_j) - y|, |a(\beta_j) - y|)$. Thus $\text{diam}(E_j) = \text{diam}(a([\alpha_j, \beta_j]))$. For a finite set of excursions, choose within each parameter interval a pair arbitrarily close to realizing that diameter. All pairs form one ordered partition of $[u, v]$; discarding the nonnegative increments between pairs and then taking finite and approximation limits gives

$$\sum_j \text{diam}(E_j)^2 \leq \|a\|_{2\text{-var};[u,v]}^2. \quad (5.2)$$

The planar isodiametric inequality gives $|\Omega_j| \leq (\pi/4) \text{diam}(E_j)^2$. Rectifiability and disjointness of the excursion intervals give

$$\sum_j \text{length}(E_j) \leq \sum_j (\text{length}(a|_{[\alpha_j, \beta_j]}) + |a(\beta_j) - a(\alpha_j)|) \leq 2 \text{length}(a),$$

so $\sum_j [E_j]$ converges absolutely in current mass. The difference $R = [L] - \sum_j [E_j]$ is a cycle supported on the chord line. The boundary is zero because boundary commutes with a mass-convergent sum when tested on differentials. Such a cycle vanishes: for a compactly supported smooth one-form, integrate its tangential coefficient along that line to obtain a smooth scalar primitive (with a transverse cutoff); the form minus its differential annihilates the tangent of R , while $R(dG) = \partial R(G) = 0$.

By (5.2) and Bieberbach's planar isodiametric inequality (Jahresbericht DMV 24 (1915), 247–250), $W = \sum_j \sigma_j [\Omega_j]$ converges in mass and $\partial W = [L]$: Interface 7, applied first to each E_j , identifies $\partial(\sigma_j [\Omega_j]) = [E_j]$, and mass convergence allows passage to the sum. The same interface gives another L^1 filling $V = [\text{Ind}(L, \cdot)]$ with the same boundary. Write $W - V = h \, dx \wedge dy$. Its boundary is zero, hence both distributional partial derivatives of h vanish. Mollification makes $h * \rho_\epsilon$ constant on $\mathbb{R}^2$; integrability forces that constant to be zero, and an approximate identity gives $h = 0$ almost everywhere. Therefore

$$\text{Ind}(L, \cdot) = \sum_j \sigma_j \mathbf{1}_{\Omega_j} \quad \text{a.e.}$$

For the cited boundary identity, given a real compactly supported test form $\omega = A \, dx + B \, dy$, take $f = A - iB$ in Cufí–Verdera and take real parts; with their $\bar{\partial}$ convention, $\text{Re}(f \, dz) = \omega$ and the real part of the area integrand is $d\omega$. This pins the orientation and sign used above. Tonelli, the triangle inequality, the isodiametric inequality, and (5.2) now prove (5.1). $\square$

Let

$$\lambda_0 = \frac{x \, dy - y \, dx}{2}, \quad d\lambda_0 = dx \wedge dy,$$

and define normalized lifted primitives

$$F_n(0) = 0, \quad F_n(t) - F_n(s) = \int_{[s,t]} c_n^* \lambda_0. \quad (5.3)$$

External interface 9 (Generalized Green formula). For every closed rectifiable planar curve L ,

$$\int_L \lambda_0 = \int_{\mathbb{R}^2} \text{Ind}(L, z) dz. \quad (5.4)$$

Here L has its displayed orientation, $dx \wedge dy$ is the positive orientation of the plane, and Ind uses the corresponding counterclockwise-positive convention.

Status: CITED: Cufí–Verdera, arXiv:1306.6832v1, main theorem, PDF p. 1. Apply it to $f(z) = \bar{z}$: since $\bar{z} dz = d(x^2 + y^2)/2 + i(x dy - y dx)$ and $\bar{\partial}\bar{z} = 1$, the exact term integrates to zero and division by $2i$ gives (5.4). Version, page, hypotheses, and sign independently checked.

Proposition 10 (Primitive compactness). *Using cited Interfaces 7 and 9, the functions F_n have a subsequence converging locally uniformly on $\mathbb{R}$.*

Status: PROVED from Lemma 6, Proposition 8, Interfaces 7 and 9, and the pinned Arzelà–Ascoli theorem.

Proof. Uniform convergence places all images in one ball $B(0, R)$. For $0 \leq t - s \leq \delta$, close $c_n|_{[s,t]}$ by its chord. Equations (5.1)–(5.4) and direct chord parametrization yield

$$|F_n(t) - F_n(s)| \leq \frac{\pi}{4} \nu(\delta)^2 + \frac{R}{2} \nu(\delta). \quad (5.5)$$

This is a common modulus tending to zero. A finite cover of $[0, 1]$ bounds F_n uniformly there, so Arzelà–Ascoli supplies uniform convergence on $[0, 1]$ along a subsequence. With $A_n = F_n(1)$,

$$F_n(k + r) = kA_n + F_n(r), \quad k \in \mathbb{Z}, \quad 0 \leq r < 1,$$

because the pulled-back integrand is periodic. This upgrades convergence to local uniform convergence on $\mathbb{R}$. $\square$

For the alternative Liouville form $\lambda_{\text{AI}} = y dx$,

$$y dx = -\lambda_0 + d(xy/2). \quad (5.6)$$

Thus its normalized primitive is

$$G_n(t) = -F_n(t) + \frac{x_n(t)y_n(t) - x_n(0)y_n(0)}{2}. \quad (5.7)$$

Equations (3.2), (5.6), and (5.7) prove local uniform convergence of G_n along the same subsequence.

Status: PROVED elementary exact-form conversion.

6 Prescribed-angle rectangle theorem

External interface 11 (Asano–Ike approximation criterion). Asano–Ike, arXiv:2412.21057v3, Theorem 1.1 (PDF p. 2), asserts that the following data:

1. a Jordan parametrization c ;
2. regular smooth Jordan parametrizations $c_n \rightarrow c$ parametrically in C^0 ; and
3. locally uniform convergence on $\mathbb{R}$ of normalized lifted primitives of the required Liouville form,

imply a prescribed-angle rectangle conclusion for every $\theta \in (0, \pi)$. The source uses $\mathbb{R}/2\pi\mathbb{Z}$ rather than $\mathbb{R}/\mathbb{Z}$; the linear change of lifted parameter preserves both kinds of convergence.

Status: CITED; exact version, theorem, page, and hypotheses independently checked against the primary PDF.

External interface 12 (Explicit nondegenerate conclusion AI_{ND}). For $\theta = \pi/2$, the conclusion imported from Asano–Ike supplies four pairwise distinct points of $c(\mathbb{R}/\mathbb{Z})$ forming a nondegenerate Euclidean rectangle whose diagonals meet at angle $\pi/2$.

Status: CITED: definition and four-distinct/off-diagonal discussion on PDF pp. 2–3, with the off-diagonal conclusion in the proof of Theorem 4.1 on PDF p. 19; independently checked against arXiv:2412.21057v3.

Lemma 13 (Perpendicular-diagonal rectangle). *A nondegenerate Euclidean rectangle whose diagonals are perpendicular is a positive-size square.*

Status: PROVED.

Proof. For adjacent side vectors u, v with $u \perp v$, the diagonal vectors are $u + v$ and $u - v$. Their scalar product is $|u|^2 - |v|^2$; perpendicularity therefore gives $|u| = |v| > 0$. $\square$

Theorem 14 (Analytic reduction to the cited rectangle criterion). *If c is a Jordan parametrization with $V_0(c)$, then c satisfies the approximation and primitive-convergence hypotheses listed in Interface 11. Consequently c has exactly the conclusion stated by the verified version of the cited Asano–Ike theorem.*

Status: PROVED using the independently verified cited interfaces.

Proof. Interface 4 and the proved interpolation estimate produce the simple polygons P_n . Lemma 5 produces regular smooth Jordan c_n satisfying (3.2). Lemma 6 gives the common local variation modulus. Proposition 10, Interface 9, and (5.6)–(5.7) give the locally uniformly convergent lifted primitives. These are precisely the data listed in Interface 11; therefore only the verified conclusion of that interface may be imported. $\square$

Corollary 15 (Positive square). *If c is a Jordan parametrization with $V_0(c)$, then c inscribes a positive-size square with four distinct vertices.*

Status: PROVED from the analytic reduction and the cited AI conclusion.

Proof. The preceding theorem verifies the hypotheses of Asano–Ike Theorem 1.1. Its independently checked four-distinct/off-diagonal conclusion at $\theta = \pi/2$, recorded in Interface 12, gives a nondegenerate rectangle with perpendicular diagonals. The perpendicular-diagonal lemma makes it a positive-size square. $\square$

7 Machine-facing dependency ledger

Component	Status	Exact interface used
Fine-mesh/local modulus	PROVED	Definition V_0 ; seam factor $\sqrt{2}$
PL interpolation error	PROVED	Every partition; constant $\sqrt{8}$
Simple PL Jordan choice	CITED	BG theorem, parameter-mesh version
Regular smoothing	PROVED	regular C^∞ embedding; (3.1)
Whole-family modulus	PROVED	tail-first then finite-prefix quantifiers
Arc winding estimate	PROVED	cited index/Jordan facts + internal excursion proof
Generalized Green	CITED	arbitrary closed rectifiable planar curve
Primitive compactness	PROVED	cited index/Green interfaces, common modulus, Arzelà–Ascoli
Liouville conversion	PROVED	equations (5.6)–(5.7)
AI approximation criterion	CITED	arXiv:2412.21057v3, Thm. 1.1, p. 2
Four distinct vertices	CITED	pp. 2–3 and Thm. 4.1 proof, p. 19
$\theta = \pi/2$ square	PROVED	cited conclusion + perpendicular-diagonal lemma

8 Pinned source identifiers

The imported interfaces have the following independently checked pins.

- Boedihardjo–Geng: arXiv:1309.1576v2, Theorem 2.2, PDF pp. 6–7, <https://arxiv.org/abs/1309.1576v2>.
- Cufí–Verdera: arXiv:1306.6832v1, main theorem, PDF p. 1 (specialized to the relevant linear test form), <https://arxiv.org/abs/1306.6832v1>.
- Carmona–Cufí, “The calculation of the L^2 -norm of the index of a plane curve and related formulae,” J. Analyse Math. 120 (2013), 225–253, DOI 10.1007/s11854-013-0019-9, preprint Theorem 2, PDF p. 8.
- Gilman–Kra–Rodríguez, *Complex Analysis: In the Spirit of Lipman Bers*, GTM 245, Springer (2007), DOI 10.1007/978-0-387-74715-6, Theorem 5.23 (Jordan curve theorem and index).
- Bieberbach, “Über eine Extremaleigenschaft des Kreises,” Jahresbericht DMV 24 (1915), 247–250 (planar isodiametric inequality).
- Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw–Hill (1976), Theorem 7.25, p. 158 (Arzelà–Ascoli).
- Asano–Ike: arXiv:2412.21057v3, Theorem 1.1, PDF p. 2; nondegeneracy discussion at PDF pp. 2–3 and the off-diagonal argument in the proof of Theorem 4.1 at PDF p. 19, <https://arxiv.org/abs/2412.21057v3>.

9 Scope

This draft neither proves the full Square Peg conjecture nor promotes PR #122 beyond **sketch**. It does not use PRs #115, #118, or #119 as premises. There is currently no problem-specific **problems/square-peg/RULES.md**; consequently the root repository rules are the only status protocol available, and this missing problem-level protocol is itself a governance gap rather than permission to promote.